



ANALYZING FINANCIAL RISK & COMPLIANCE TRENDS

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The Problem Statement

Financial risk assessment involves identifying and analyzing risks related to fraud, compliance violations, and auditing failures.

The Big 4 consulting firms – Ernst & Young (EY), PwC, Deloitte, and KPMG – play a crucial role in financial risk assessment by conducting audits, ensuring compliance, and advising businesses on risk management.

- They help companies mitigate market, credit, and compliance risks.
- They utilize AI and data analytics for fraud detection and trend analysis.
- They assess client financial health to provide strategic insights.

Objective

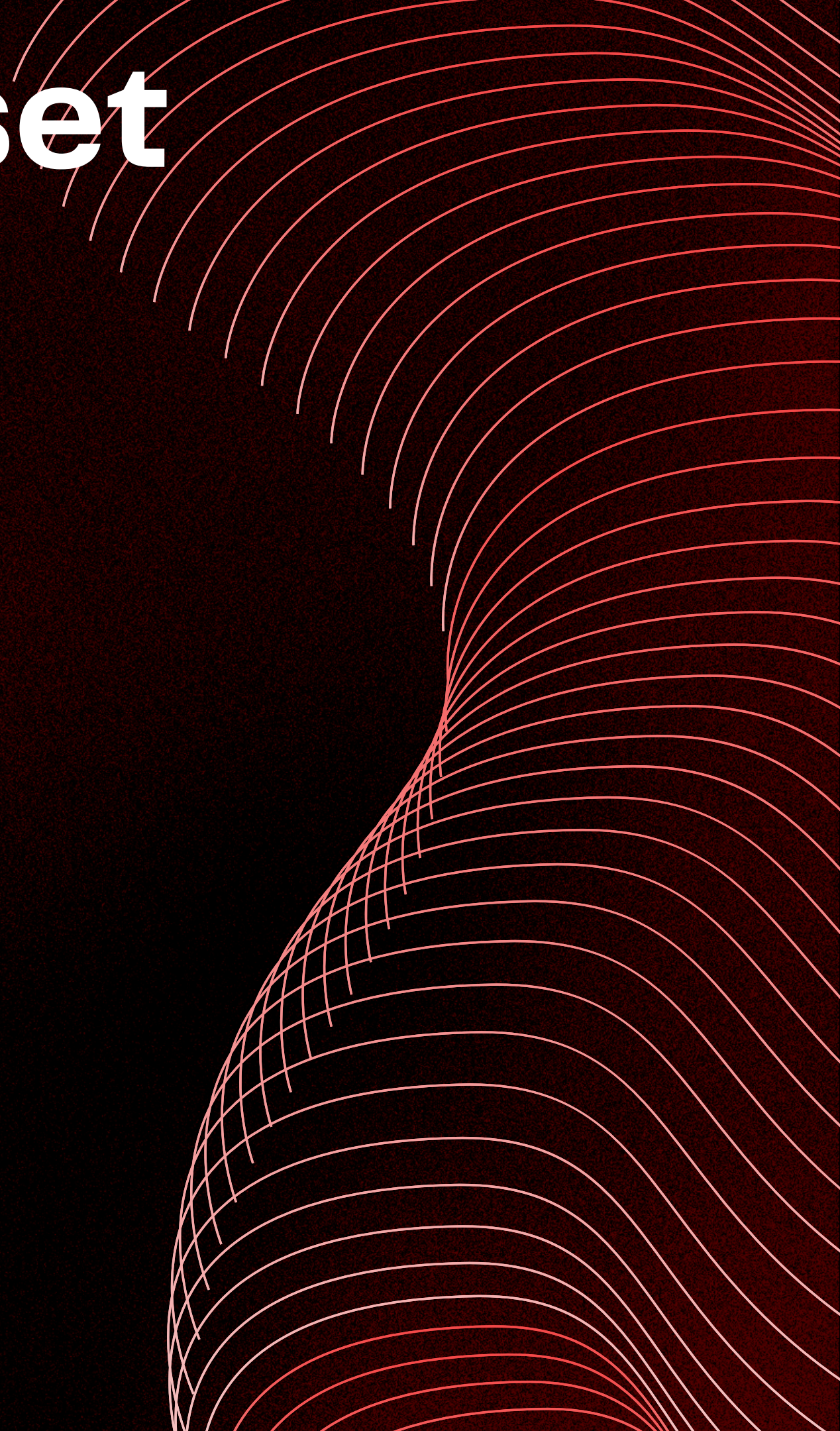
Modern financial risk assessment has been revolutionized by Artificial Intelligence (AI), which provides:

- Automated Risk Detection – AI models analyze large datasets for anomalies.
- Real-time Fraud Monitoring – AI-driven fraud detection systems flag suspicious transactions instantly.
- Predictive Analysis – Machine learning models forecast potential financial risks.

Objective of the Study

- To analyze audit effectiveness using Randomized Complete Block Design (RCBD) and Factorial Experiments on real-world data.
- To explore the impact of AI in auditing and employee workload on audit effectiveness.

Key Variables in Dataset

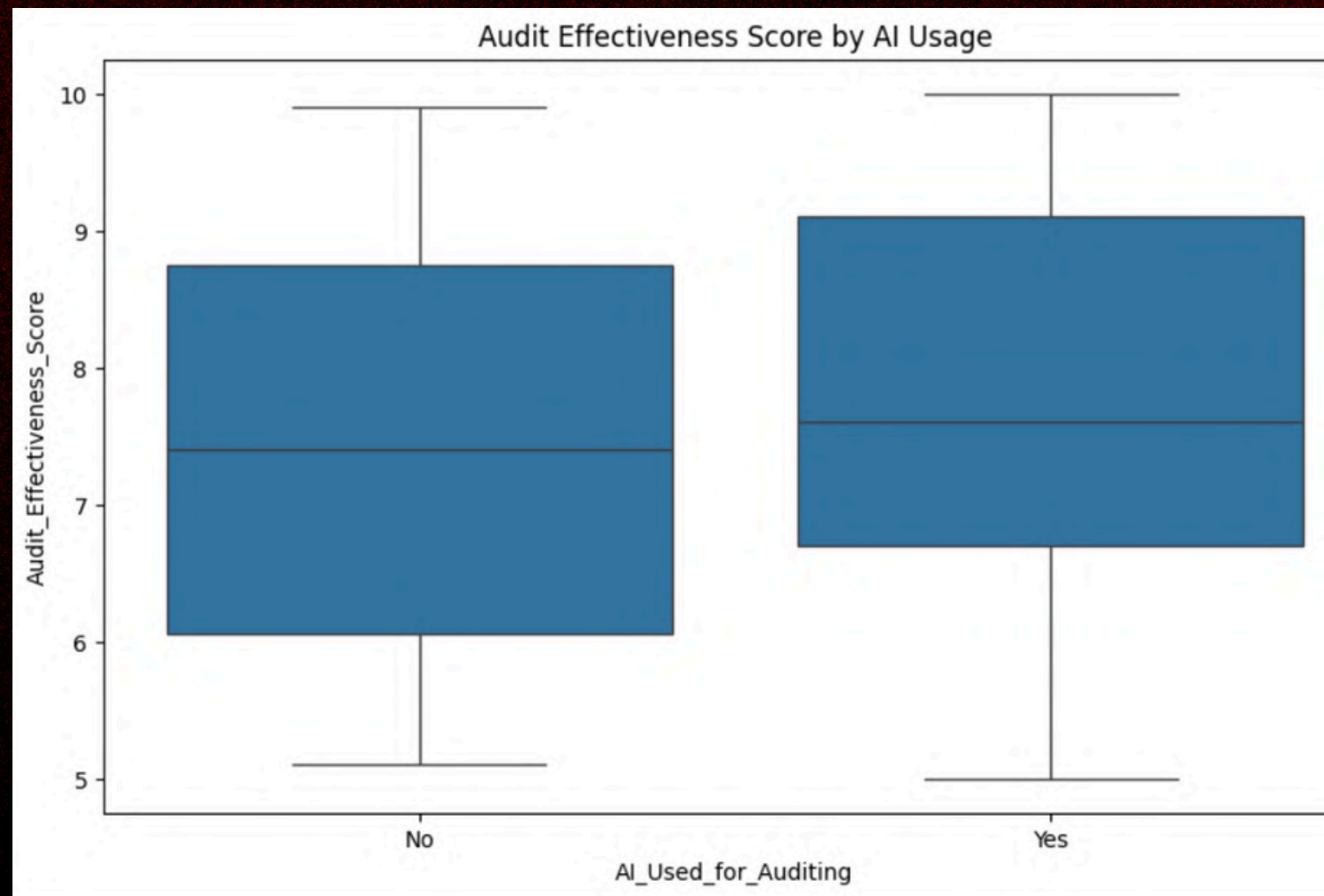


- ***Audit Engagements***
- ***High-Risk Cases***
- ***Fraud Cases Detected***
- ***AI Usage in Auditing***
- ***Audit Effectiveness Score***

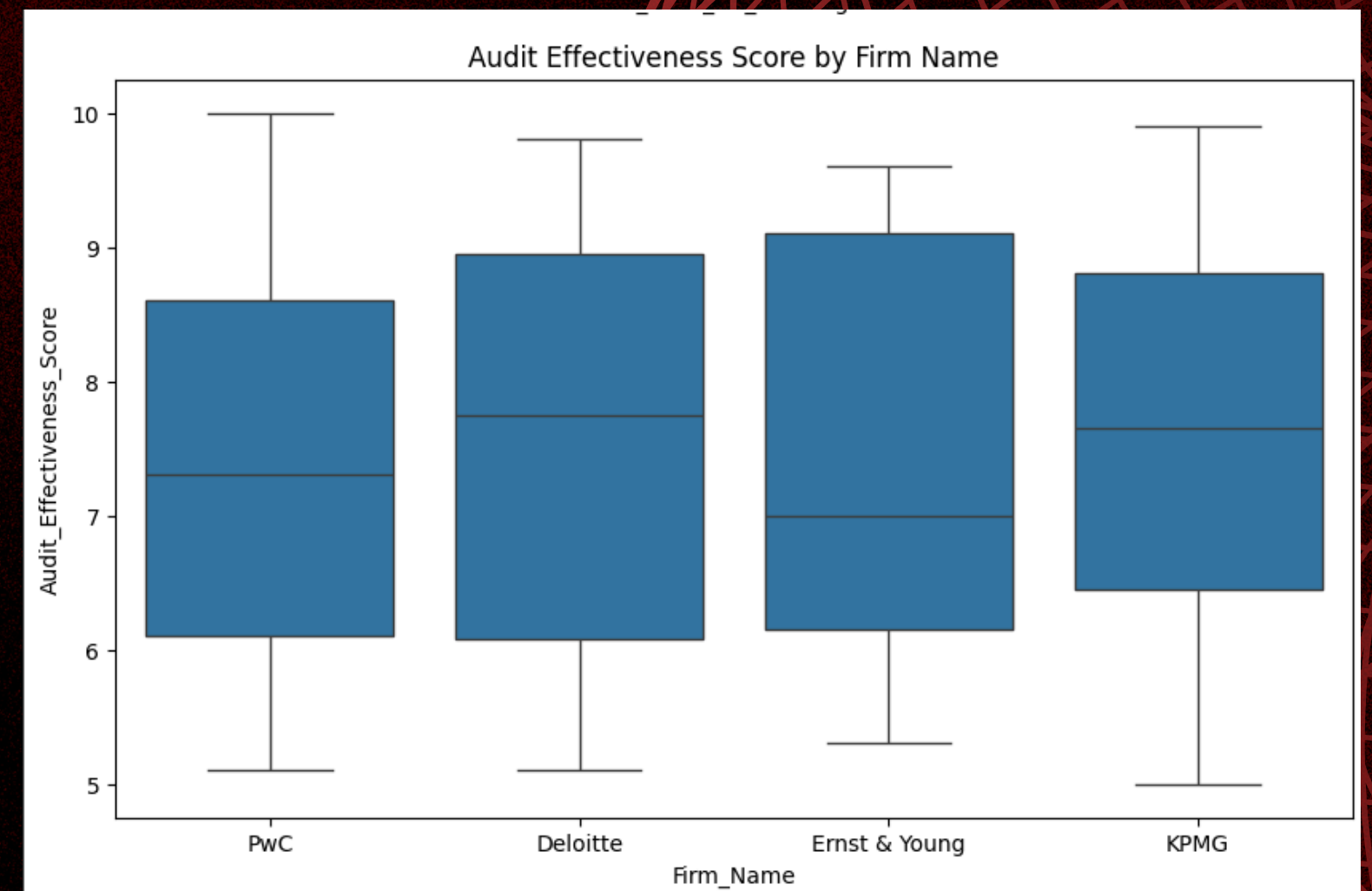
Data Visualization

Boxplots of Audit Effectiveness Score:

By AI Usage (Yes/No)



By Firm Name



Why not choose Latin Square Design?

What is Latin Square Design?

- Used when two blocking variables (e.g., rows and columns) influence the response.
- Each treatment appears exactly once in each row and column.
- Assumes a square layout (same number of rows, columns, and treatments).

Why We Did Not Use It

- Our dataset lacks a second structured blocking variable besides the firm (like time or auditor).
- Treatment structure (e.g., AI usage) does not align with a square arrangement.
- The real-world audit data doesn't follow the rigid layout Latin Square requires.

Why not choose BIBD ?

What is BIBD?

- An incomplete block design where:
 - Not all treatments appear in each block.
 - Each pair of treatments appears together in a block a fixed number of times.
- Used when testing many treatments, but blocks can accommodate only a few.

Why We Did Not Use It

- We had only 2–3 treatment levels (e.g., AI: Yes/No; Workload: Low/Med/High).
- All treatments are fully observed in all firms — so blocks are complete, not incomplete.
- No need for balancing based on treatment co-occurrence.

Why We Chose RCBD and Factorial Design ?

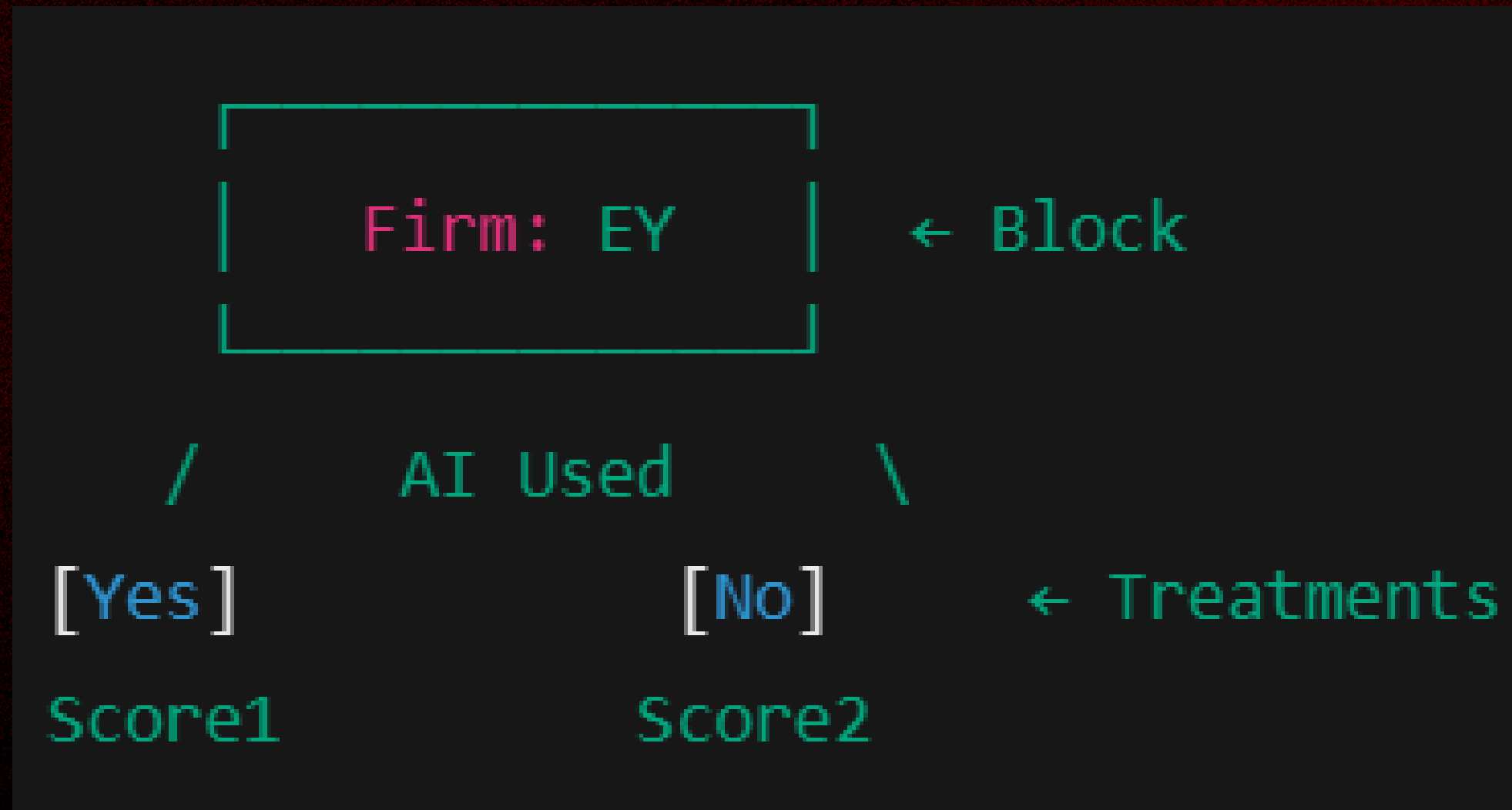
Why RCBD?

- Audit effectiveness might vary across firms (EY, PwC, etc.), independent of AI usage.
- Blocking helps remove firm-specific variation, isolating AI's effect.

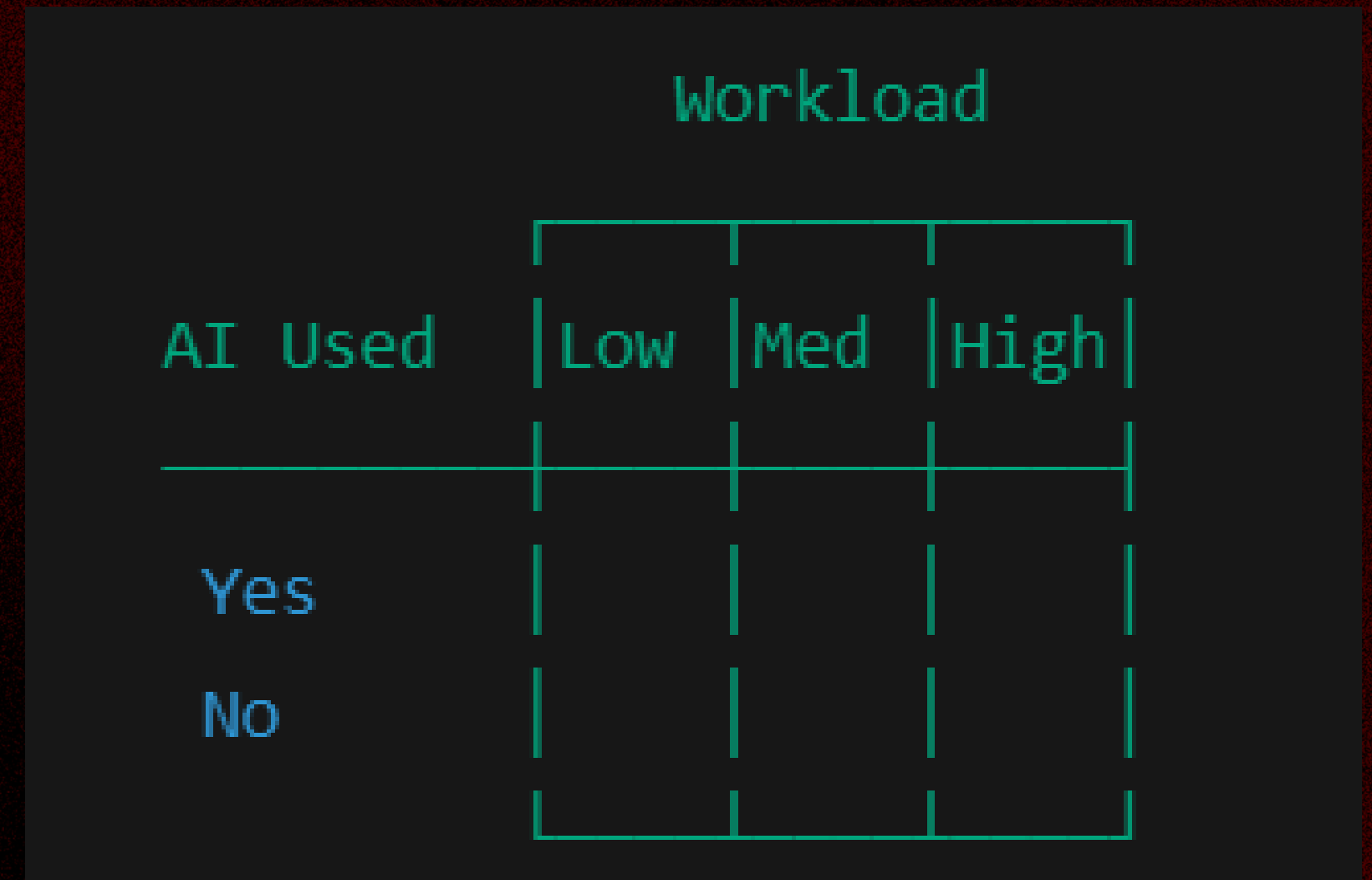
Why Factorial?

- We suspected that AI impact may depend on workload level.
- This design helps understand main effects + interaction.

Why We Chose RCBD and Factorial Design ?



RCBD



Factorial Design

Randomized Complete Block Design (RCBD)

Objective:

To assess the effect of AI in auditing while controlling for firm-level differences.

Experimental Design:

- Factor: AI Usage (Yes/No)
- Blocking Factor: Firm Name (EY, PwC, Deloitte, KPMG)
- Response Variable: Audit Effectiveness Score

Hypotheses:

- *Null Hypothesis (H_0): AI usage doesn't have significant impact on audit effectiveness.*
- *Alternative Hypothesis (H_1): AI usage has a significant impact audit effectiveness.*

Randomized Complete Block Design (RCBD) ANOVA Table

Source	SS	DF	MS	F
AI Usage	0.9116161616161629	1	0.9116161616161629	0.3829628081715222
Firm Name	0.5375328063241054	3	0.17917760210803513	
Error	226.14085103205974	95	2.380430010863787	
Total	227.59	99		

F-statistic (AI Usage): 0.3830
F-critical ($\alpha = 0.05$): 3.9412
Hypothesis: Accept the Null Hypothesis.

Factorial Design



Objective:

- To analyze the combined impact of AI usage and employee workload on audit effectiveness.

Experimental Design:

- Factor 1: AI Usage (Yes/No)
- Factor 2: Employee Workload (Low/Medium/High)
- Response Variable: Audit Effectiveness Score

Hypotheses:

- H_0 (Main Effect 1): AI has significant effect on audit effectiveness.
- H_0 (Main Effect 2): Workload has significant effect on audit effectiveness.
- H_0 (Interaction Effect): Interaction exists between AI and workload.

Factorial Design ANOVA Table

Source	SS	DF	MS	F
AI Usage	0.9116161616161629	1	0.9116161616161629	0.408648139499395
Workload	5.282532467532452	2	2.641266233766226	1.1839945119419717
Interaction	11.699756101962546	2	5.849878050981273	2.6223117606798034
Error	209.69609526888883	94	2.2308095241371153	
Total	227.59	99		
F-statistic (AI Usage): 0.4086, F-critical: 3.9423 → Accepted H ₀ (AI effect)				
F-statistic (Workload): 1.1840, F-critical: 3.0933 → Accepted H ₀ (Workload effect)				
F-statistic (Interaction): 2.6223, F-critical: 3.0933 → Accepted H ₀ (Interaction effect)				

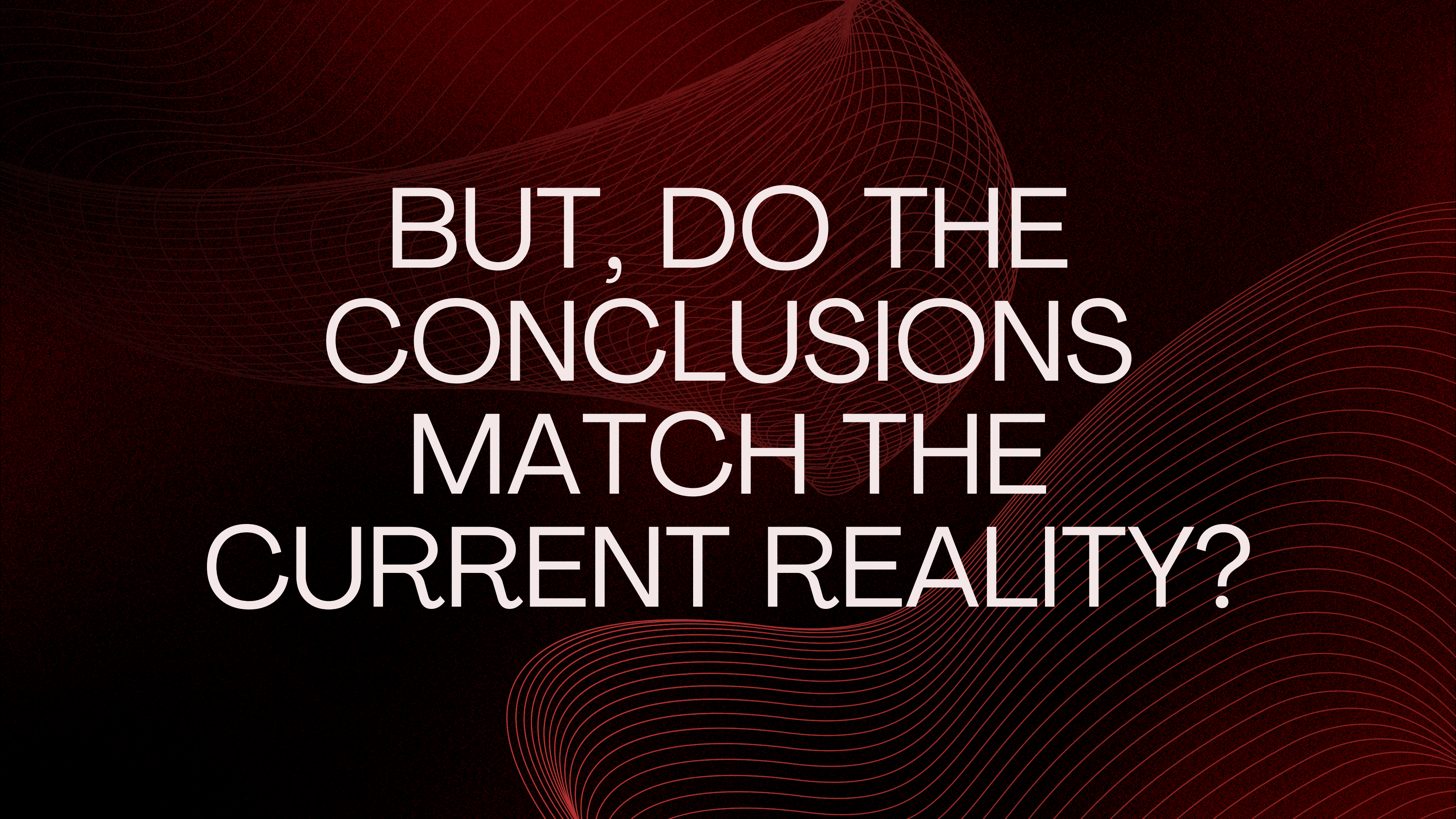
Key Findings

RCBD Findings:

- AI usage does not influence audit effectiveness, even after controlling for firm-level differences. However, the data is too short to comment on AI's effect.

Factorial Findings:

- AI usage does not improve effectiveness overall.
- Also, there is no significant correlation between the workload and effectiveness.
- No significant interaction/correlation between AI and workload.



BUT, DO THE
CONCLUSIONS
MATCH THE
CURRENT REALITY?

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THANK YOU